NAME : KOWSHIKA I R
ROLL NO : 231501080
EX NO : 5

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

# Load dataset
df = pd.read_csv("CC GENERAL - CC GENERAL (1).csv")

# Remove ID column first
df.drop("CUST_ID", axis=1, inplace=True)

# Handle missing values (only numeric columns)
df.fillna(df.mean(numeric_only=True), inplace=True)

# Scaling
scaler = StandardScaler()
X = scaler.fit_transform(df)

# Elbow Method
wcss = []
for i in range(1,11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

plt.plot(range(1,11), wcss)
plt.xlabel("Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()

# Apply KMeans (choose elbow value, usually 4)
kmeans = KMeans(n_clusters=4, random_state=42)
df["Cluster"] = kmeans.fit_predict(X)

# Silhouette Score
print("Silhouette Score:", silhouette_score(X, df["Cluster"]))

# Cluster Profiling
print(df.groupby("Cluster").mean())
```

paps ex 2,3,4.ipynb

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sample_data

CC GENERAL - CC GENERAL (1)....

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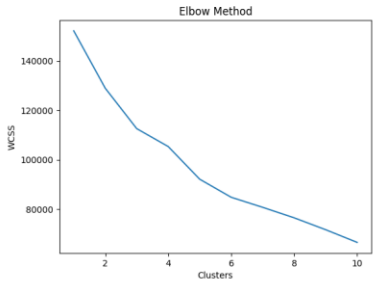
```
print(df.groupby("Cluster").mean())
```




```
Silhouette Score: 0.16639208656701637
              BALANCE  BALANCE_FREQUENCY    PURCHASES  ONEOFF_PURCHASES  \
Cluster
0        5023.958508           0.967107   721.184928        453.485000
1         109.464707           0.412883   351.589913        191.487250
2        1253.422099           0.974766  2181.418772       1241.370362
3        1500.348531           0.966640   282.216074        209.226223

         INSTALLMENTS_PURCHASES  CASH_ADVANCE  PURCHASES_FREQUENCY  \
Cluster
0                    267.804742   5075.737329             0.329511
1                    160.571811    299.684391             0.320703
2                    940.311888    234.489319             0.928436
3                     73.305481    789.088912             0.206247

         ONEOFF_PURCHASES_FREQUENCY  PURCHASES_INSTALLMENTS_FREQUENCY  \
Cluster
```

Variables Terminal

11:01 AM Python 3